Global 1-km Climatic Design Extremes at Ungauged Locations via Machine Learning Regionalization of Extreme Value Parameters

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ABSTRACT

Accurate climatic design data are essential for HVAC sizing, building envelope evaluation, and resilience planning, particularly for rare but consequential extremes such as the 1% annual exceedance probability (AEP) dry-bulb temperature. Many locations lack sufficiently long observational records to estimate such return levels directly, while coarse reanalysis products do not reliably represent point-scale extremes. I present a global, observation-informed framework for estimating annual maximum dry-bulb return levels at 30 arc-seconds ($\sim$1 km) resolution. The approach combines two stages: station-level Bayesian non-stationary Gumbel fits to 23,935 quality-controlled Global Historical Climatology Network records, followed by XGBoost regionalization of the fitted parameters using 19 physically interpretable climatic and topographic predictors. Across spatially blocked holdout stations, the framework reproduces the 1% AEP return level with $R^2 = 0.92$ and MAE $= 1.02$°C (1.84°F), reducing MAE by roughly two-thirds relative to a direct ERA5-based Gumbel benchmark [$R^2 = 0.39$, MAE $= 2.86$°C (5.15°F)] and eliminating its $\sim$2.2°C (4.0°F) cold bias. The resulting dataset provides spatially continuous, observation-aligned design extremes for ungauged locations worldwide, with the largest practical value in data-sparse regions such as sub-Saharan Africa, interior South America, and high-latitude interior Eurasia, and in areas with short or noisy station records. The framework is directly extensible to other climatic design variables, including annual minimum dry-bulb temperature, humidity extremes, and heating and cooling degree days.

Keywords: climatic design conditions; extreme value analysis; Gumbel distribution; XGBoost; regionalization; GHCN; ERA5; dry-bulb temperature

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1. Introduction

Accurate climatic design information underpins HVAC sizing, building envelope performance, and resilience planning. Recent extremes illustrate how high-consequence events can approach or exceed conventional design thresholds: during the February 2021 cold outbreak in Texas, temperatures in Dallas fell to approximately -19°C (-2°F), while the June 2021 Pacific Northwest heat dome brought Portland to 47°C (116°F) (The Perryman Group 2021; Hobbs and Stein 2021; White et al. 2023; Oregon Health Authority 2023). These events underscore the value of characterizing not only common design conditions such as 10%, 5%, and 2% annual exceedance probability (AEP) levels, but also rarer extremes such as the 1% AEP return level.

In data-sparse areas, practitioners often rely on the nearest weather station or coarse atmospheric reanalysis products such as ERA5 (Hersbach et al. 2020). Reanalysis provides complete global coverage, but its approximately 30–50 km resolution and grid-cell averaging smooth local microclimates and often understate point-scale thermal extremes (Smith et al. 2025; Sheridan et al. 2020; Lopes et al. 2024). For engineering applications, that scale mismatch can bias equipment sizing and envelope specifications (Roth 2020).

To address this gap, I developed a global framework for estimating annual maximum dry-bulb return levels at 30 arc-seconds ($\sim$1 km) resolution. The approach combines station-level non-stationary Gumbel modeling with machine-learning regionalization of the resulting parameters, preserving observation-based extreme-value behavior while extending estimates to ungauged locations. The method builds on recent advances in regionalized extreme estimation to produce spatially complete data for observed and unobserved locations worldwide (Hoch et al. 2025).

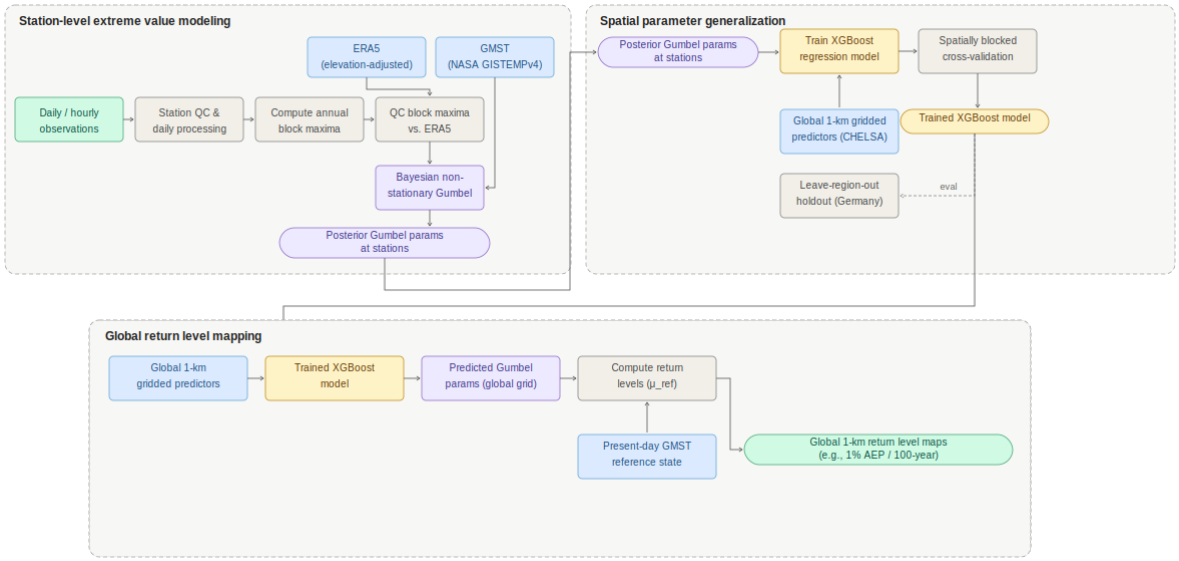


Figure 1. Overview of the three-stage modeling pipeline: (1) station-level non-stationary Gumbel extreme value modeling, (2) spatial parameter generalization via XGBoost, and (3) global return level mapping at 1-km resolution.

2. Data and Methods

Figure 1 summarizes the three-stage workflow, beginning with station-level non-stationary Gumbel fits to quality-controlled observations, followed by XGBoost-based regionalization of posterior mean parameters using global 1-km predictor layers, and ending with spatially continuous mapping of present-day return levels.

2.1. Observations

Daily maximum surface air temperatures were obtained from the Global Historical Climatology Network daily dataset (GHCNd) (Menne et al. 2012) and, where available, the Global Historical Climatology Network hourly dataset (GHCNh) (Menne et al. 2024) for the period 1940–2024. For GHCNh stations, daily maxima were computed as the highest hourly observation within each calendar day, excluding days with fewer than 18 valid hourly values. Although GHCN data undergo standard quality control, additional screening was applied following the HadISD quality control framework developed for synoptic station data by Dunn et al. (2012, 2016), including removal of duplicate records, physically implausible values, and observations failing internal consistency checks. Annual extrema were derived from daily series, retaining only years with at least 85% valid days per month and at least 10 sufficiently complete months. Annual maximum temperature was computed for qualifying years.

Annual station extremes were further screened against elevation-adjusted ERA5 annual extremes at corresponding locations and years. ERA5 values were interpolated to station coordinates and adjusted for the elevation difference between the ERA5 grid cell and the station elevation using a standard atmospheric lapse rate of 6.5°C km^{-1}. Outliers were identified using a robust mean absolute deviation–based z-score together with an absolute residual threshold of ±5°C (±9°F). This conservative procedure prioritizes physical consistency for extreme value modeling, though it may occasionally remove legitimate extremes where local conditions diverge substantially from reanalysis representation. Stations with fewer than 20 valid annual values were excluded. In total, 23,935 GHCN stations were retained.

2.2. Non-stationary Gumbel Extreme Value Model

Extreme Value Framework. Annual maximum temperatures at each station are modeled as independent realizations from a Gumbel distribution (Coles 2001), a member of the generalized extreme value family. For annual maxima, the right-skewed Gumbel is applied. The Gumbel has a two-parameter structure: a location parameter μ and a scale parameter $\beta > 0$. The location parameter describes the typical level of annual extreme temperatures, while the scale parameter controls how much those extreme values vary from year to year. The log-likelihood contribution of a single annual extreme y_t is:

$$\ell(\mu, \beta \mid y_t) = -\log\beta - z_t - \exp(-z_t) \qquad (1)$$

where $z_t = (y_t - \mu_t)/\beta$. The Generalized Extreme Value formulation was also evaluated; however, the estimated shape parameter ξ was very close to zero across the global station network with a mean of −0.005 and median of −0.001, and 95% of values falling within ±0.04 of zero. The more parsimonious Gumbel model ($\xi = 0$) was therefore adopted.

Non-stationary Formulation. Global mean temperature has risen steadily under anthropogenic forcing, and thermodynamic theory predicts a corresponding shift in the distribution of local temperature extremes in both tails (Fischer and Knutti 2015; Seneviratne et al. 2021). Whether this global signal has produced statistically detectable trends in local extreme dry-bulb temperatures at individual stations, particularly in annual block maxima, remains an open empirical question, as station-level trend detection is complicated by high interannual variability, sparse and inhomogeneous records, and limited spatial coherence in extremes indices (Dunn et al. 2020). Nevertheless, I capture any changes in local annual maxima by allowing the location parameter to vary linearly with global mean surface temperature (GMST) anomaly, ΔT, a physical index of the externally forced state of the climate system:

$$\mu_t = \mu_0 + \mu_1 \cdot x_t \tag{2}$$

where μ_0 is the baseline location and μ_1 is the sensitivity of the extreme distribution to global warming. For annual maxima, a positive μ_1 indicates intensifying heat extremes. The covariate x_t is the GMST anomaly standardized to zero mean and unit variance over the station record:

$$x_t = \frac{\Delta T_t - \overline{\Delta T}}{s_{\Delta T}} \tag{3}$$

where $\overline{\Delta T}$ and $s_{\Delta T}$ are the mean and standard deviation of the GMST anomaly over the station's period of record, respectively. This standardization renders μ_0 interpretable as the location parameter at the historical mean climate state, and places μ_1 on a comparable scale across stations. The scale parameter β is held constant in time. GMST data are obtained from the NASA GISS Surface Temperature Analysis, version 4 (Lenssen et al. 2024; GISTEMP Team 2024). All anomalies are expressed relative to the 1951–1980 base period.

Bayesian Parameter Estimation. Parameters $\theta = \{\mu_0, \mu_1, \beta\}$ are estimated within a Bayesian framework. I assign independent, weakly informative priors to each parameter. For the location parameters, specified on the normalized observation scale,

$$\mu_0 \sim \text{Normal}(0,\, 1)$$
$$\mu_1 \sim \text{Normal}(0,\, 0.5)$$

The tighter prior on μ_1 reflects the expectation that the sensitivity of local extremes to global mean warming is modest relative to the overall spread of the distribution, while still permitting the range of physically plausible responses. For the scale parameter, a log-normal prior is used to enforce positivity:

$$\log \beta \sim \text{Normal}(0,\, 0.5)$$

Posterior inference was performed using the No-U-Turn Sampler (Hoffman and Gelman 2014) with four independent chains. Convergence was confirmed for all station fits using the potential scale reduction factor ($\widehat{R} \leq 1.01$) and bulk and tail effective sample size diagnostics (Vehtari et al. 2021).

Return Level Estimation. The T-year return level is the temperature exceeded with annual probability $1/T$. Under the Gumbel model, the return level at a reference climate state is:

$$\text{RL}(T) = \mu_{\text{ref}} + \beta \cdot \log\left[-\log\left(1 - \frac{1}{T}\right)\right] \tag{4}$$

Return levels are conditioned on a present-day reference state defined as the mean GMST anomaly over the most recent 20 years of each station's record, giving a reference location $\mu_{\text{ref}} = \mu_0 + \mu_1 x_{\text{ref}}$.

2.3. Global Feature Data

To regionalize the spatially sparse Gumbel parameters, I assembled a set of global predictor layers representing the primary geographic and climatic controls on extreme temperature, including topography, mean climate state, seasonality, moisture availability, surface energy balance, and synoptic-scale circulation influences (Table 1).

2.4. Machine Learning Framework

I use Extreme Gradient Boosting (XGBoost), a machine learning method that sequentially combines decision trees to minimize residual error (Chen and Guestrin 2016). Regularization parameters help limit overfitting, which is especially important here given the spatial clustering of stations and noise in station-level extreme value parameters.

The model is trained to predict Gumbel distribution parameters across a spatial grid by learning nonlinear relationships and sharp gradients from the geographic and climatic predictors listed in Table 1. Categorical variables are one-hot encoded, and the scale parameter (β) is modeled in log-space to enforce positivity and improve numerical stability. Modeling parameters directly allows any desired return level to be derived analytically while preserving the statistical structure of extremes.

Hyperparameters are selected via a randomized search of 40 configurations, evaluated using five-fold spatially blocked cross-validation with R^2 as the primary criterion. To account for spatial autocorrelation, stations are assigned to $2^\circ \times 2^\circ$ geographic blocks; all cross-validation and train-test splitting is conducted at the block level. A spatially blocked holdout comprising 20% of these blocks is withheld entirely from training to provide a realistic assessment of predictive skill at ungauged locations.

2.5. Model Evaluation

Predictive skill for each Gumbel parameter is quantified using the coefficient of determination (R^2), mean absolute

Table 1. Overview of predictor layers used. Unless otherwise noted, source = CHELSA v2.1 and native resolution = 30 arc-seconds.

Gridded Predictor	Source	Native Res.
Latitude	—	—
Elevation	GMTED2010	30 arc-sec
Mean sea level pressure	ERA5	0.25°
Annual mean temperature	CHELSA v2.1	30 arc-sec
Mean diurnal temperature range		
Day-to-night vs. summer-to-winter temperature variation		
Temperature seasonality (standard deviation)		
Maximum temperature of warmest month		
Temperature annual range		
Mean temperature of warmest quarter		
Precipitation seasonality (coefficient of variation)		
Precipitation of warmest quarter		
Mean cloud fraction		
Mean vapor pressure deficit		
Mean surface wind speed		
Maximum downwelling shortwave radiation		
Mean of annual maximum temperatures		
Maximum of annual maximum temperatures		
Coefficient of variation of annual maximum temperatures		

error (MAE), and root mean square error (RMSE). Beyond global metrics, I assess bias in regionalized parameters and the 1% AEP using percentage bias (PBIAS):

$$\text{PBIAS}\,[\%] = \frac{(A-B)}{B} \times 100 \tag{5}$$

where A represents the regionalized estimate and B the station-derived benchmark. PBIAS is summarized globally and by region to identify geographically coherent patterns of under- or overprediction.

Finally, to evaluate generalization in data-sparse regions, I conduct a "leave-region-out" experiment by excluding all stations within Germany from training. I refit the model and evaluate its performance at these "unseen" locations against independent station-level estimates. Comparing these results to the full-dataset model quantifies the extent to which predictions rely on transferable environmental relationships rather than spatial proximity to training data.

Table 2. Mean evaluation scores of the best-performing set of hyperparameters per Gumbel parameter across a 5-fold cross-validation.

	Location (μ)	Scale (β)
R^2 [–]	0.93	0.60
MAE [°C]	0.94 (1.69°F)	0.21 (0.38°F)
RMSE [°C]	1.32 (2.38°F)	0.28 (0.50°F)

Table 3. Cross-validation skill for the 1% AEP return level (RL_{100}) across all holdout stations.

Metric	RL_{100}
R^2 [–]	0.92
MAE [°C]	1.02 (1.84°F)
RMSE [°C]	1.39 (2.50°F)

3. Results and Discussion

3.1. Machine Learning Model Accuracy

Table 2 summarizes the evaluation scores for the best-performing hyperparameter configuration. Predictive skill is substantially higher for the location parameter than for the scale parameter. Approximately 93% of the spatial variation in the location parameter is explained by the model, compared to 60% for the scale parameter. This indicates that geographic and climatological predictors capture the large-scale structure of the location parameter more effectively than they capture the scale parameter. The comparatively lower skill for the scale parameter suggests that additional predictors representing year-to-year variability in extreme temperatures may be required to further improve performance.

The lower predictive skill for β raises the question of how scale errors affect return level accuracy. Under the Gumbel model, $RL_{100} = \mu + 4.6 \times \beta$, so errors in β are amplified in the 100-year return level. At the median station, a $\pm 25\%$ error in β would translate to roughly ± 1.8°C (± 3.2°F) in RL_{100}. Importantly, however, β PBIAS is centered near zero globally (Figure 3), indicating scatter rather than systematic bias. Consistent with that result, RL_{100} PBIAS is also centered near zero, suggesting that uncertainty in β increases site-level noise but does not introduce a directional bias in regionalized return levels.

Cross-validation skill was therefore evaluated directly for the 1% AEP return level (RL_{100}) computed from the predicted Gumbel parameters (Table 3). The R^2 of 0.92 closely matches the skill for μ, indicating that the broad spatial pattern of rare heat extremes is well reproduced. The MAE of ~ 1°C (~ 1.8°F) is also consistent with the scale-sensitivity calculation above, suggesting that the regionalized parameters capture most of the station-level signal despite lower skill in β.

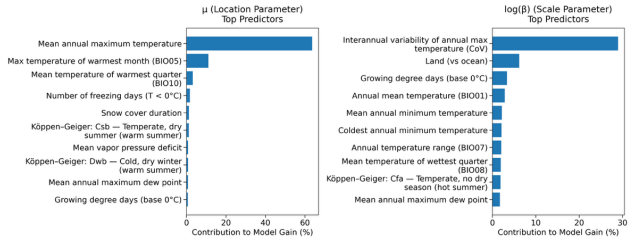


Figure 2. Top predictors of Gumbel distribution parameters by percentage contribution to model gain. Mean annual maximum temperature dominates the location parameter (μ), while interannual variability of annual maximum temperature dominates the scale parameter [$\log(\beta)$].

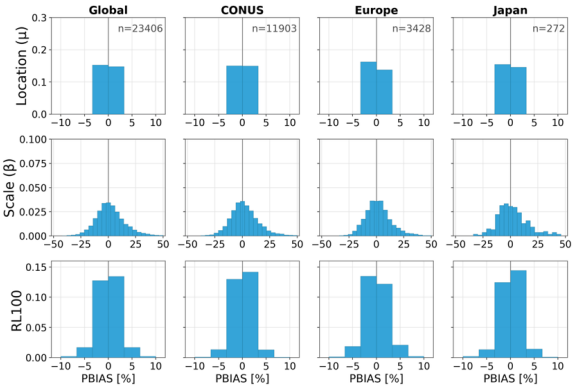


Figure 3. Comparison of percentage bias (PBIAS) for Gumbel location (μ), scale (β), and the 1% annual exceedance probability return level (RL_{100}) across GHCN stations. Panels show the distribution of bias globally and by region. Zero PBIAS indicates exact agreement between local and regionalized estimates; negative values signify model underprediction.

Feature importance is physically coherent. For μ, mean annual maximum temperature dominates the model gain, indicating that the regional thermal baseline largely sets the center of the extreme value distribution. For $\log(\beta)$, the leading predictor is the interannual variability of annual maximum temperature, consistent with β representing year-to-year volatility in extremes. This separation is encouraging because it suggests the model is learning distinct controls on typical extreme intensity versus dispersion.

The regionalized model also performs well at station locations, with PBIAS distributions centered near zero for μ, β, and RL_{100}. Bias in μ is tightly constrained globally, while β shows a broader but still well-balanced spread. Similar behavior across CONUS, Europe, and Japan indicates that the framework preserves the statistical character of point-based observations when translated into a continuous field.

3.2. Model Skill in Regionalizing to Ungauged Areas

To evaluate out-of-region generalization, I withheld all GHCN stations within a bounding box encompassing Germany (5°E–16°E, 47°N–55.5°N) from both model training and hyperparameter optimization. Evaluated at the 979 German stations, skill for the 1% AEP return level declined from $R^2 = 0.80$ for the full-data model (which included German stations in training) to $R^2 = 0.69$ for the Germany-excluded model (Table 4), with MAE increasing from 0.98°C (1.77°F) to 1.26°C (2.28°F). The lower full-data skill over Germany relative to the global holdout value ($R^2 = 0.92$; Table 3) reflects the narrower dynamic range of extremes within a single mid-latitude country, which compresses explained variance. The modest drop when German stations are excluded suggests the framework can transfer environmental relationships into ungauged areas, while still benefiting from local observations where available.

Table 4. Performance comparison for 1% AEP predictions within Germany, contrasting models trained with and without stations located within Germany.

	With Germany	No Germany
R^2 [–]	0.80	0.69
MAE [°C]	0.98 (1.77°F)	1.26 (2.28°F)
RMSE [°C]	1.27 (2.30°F)	1.59 (2.86°F)

3.3. Regionalized Return Level Intensities

Figure 4 compares station-based 1% AEP estimates of annual maximum dry-bulb temperature with the regionalized field. Dense station support across North America, Europe, and parts of East Asia contrasts with major gaps in central Africa, interior South America, and high-latitude interior Eurasia. The machine-learned map reproduces the broad structure visible in well-observed regions while extending estimates continuously across these gaps.

A direct comparison against an ERA5-based Gumbel benchmark further highlights the value of the regionalized framework (Table 5). Across all holdout stations, the ML model achieves $R^2 = 0.92$ versus 0.39 for the ERA5 Gumbel, with MAE of 1.02°C (1.84°F) versus 2.86°C (5.15°F) and bias of −0.02°C (−0.04°F) versus −2.24°C (−4.03°F). The ERA5 benchmark systematically underestimates RL_{100} by about 2.2°C (4.0°F), consistent with the smoothing of local thermal extremes in ∼30-km reanalysis grid cells (Smith et al. 2025; Sheridan et al. 2020; Lopes et al. 2024). For HVAC applications, that underestimation would imply a meaningful underestimate of peak cooling load, whereas the observation-informed framework largely corrects this scale mismatch.

4. Conclusion and Future Work

This study presents a global, observation-informed framework for estimating rare dry-bulb temperature return levels at ungauged locations. The framework shows strong cross-validated skill for RL_{100}, maintains $R^2 = 0.69$ in the Ger-

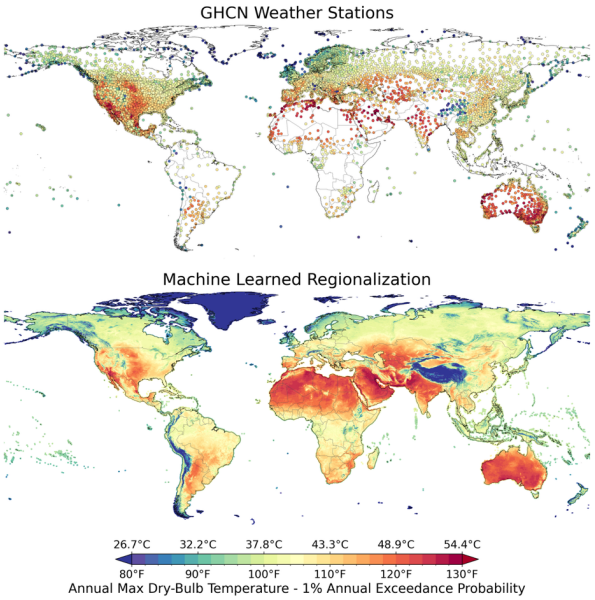


Figure 4. Global distribution of the 1% annual exceedance probability (100-year return level) for annual maximum dry-bulb temperature. The upper panel displays local estimates derived from the Global Historical Climatology Network (GHCN) stations. The lower panel shows the regionalized output produced by the machine learning model, providing spatially complete coverage at 30 arc-seconds (∼1-km) resolution.

Table 5. Comparison of ML model and ERA5 Gumbel approach for 1% AEP return level (RL_{100}) prediction across all holdout stations.

Metric	ML model	ERA5 Gumbel
R^2 [–]	0.92	0.39
MAE [°C]	1.02 (1.84°F)	2.86 (5.15°F)
RMSE [°C]	1.39 (2.50°F)	3.88 (6.98°F)
Bias [°C]	−0.02 (−0.04°F)	−2.24 (−4.03°F)

many holdout test, and substantially outperforms a direct ERA5-based Gumbel benchmark (Table 5).

By recovering point-scale design extremes more faithfully than raw reanalysis, the method helps fill a persistent gap in climatic design data for places where station records are short or sparse, supporting HVAC sizing, building envelope specification, and resilience planning.

The framework also adds value in areas where short or noisy station records yield unstable return-level estimates at nearby sites within otherwise homogeneous climates. In such cases, individual station fits can produce physically implausible contrasts in return levels over small distances; the regionalization step pulls these noisy estimates toward the pattern of their climatic neighbors, yielding a more spatially coherent field while preserving observation-based information.

Two limitations merit emphasis. First, the scale parameter is harder to predict than the location parameter, and its uncertainty propagates into return levels; additional predictors of interannual variability and full uncertainty propagation would improve the product. Second, out-of-region generalization was tested only for Germany, which is a relatively favorable holdout because it has dense station coverage and is surrounded by well-observed regions. Performance in more weakly sampled regions such as sub-Saharan Africa, interior South America, and central Asia may be more variable. Additional sparse-region holdouts are needed to better characterize extrapolative performance.

A related caveat applies to the skill metrics themselves: reported R^2, MAE, and bias values compare regionalized predictions against station-level posterior mean return levels, which are themselves estimates with non-trivial sampling uncertainty rather than a single "true" value. The posterior credible interval on a 100-year return level fit from a typical record can be substantial, and this irreducible uncertainty bounds how closely any regionalization, ML-based or otherwise, can match point estimates. The same uncertainty applies to any station-derived return level used in engineering practice, whether or not it is explicitly propagated in design calculations.

This framework is readily extensible to annual minimum dry-bulb temperature, peak wet-bulb temperature, and cooling and heating degree days, each of which is currently constrained to locations with sufficient station records. Extending this approach across the full suite of climatic design variables would provide a comprehensive, observation-aligned global resource to support building design and infrastructure resilience in areas without long-term, high-quality in situ observations.

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